Bogleheads

Approximating total international stock market

> *This article contains details **specific to United States (US) investors**. If you are a non-US investor, **do not use** any funds or ETFs listed here. They expose you to harmful US dividend tax and estate tax risks. Proceed **only** if you have verified these risks do not apply to you.*

Approximating total international stock market shows how you can combine funds to mimic the composition of the international stock market. Funds can be mutual funds, index funds or exchange-traded funds (ETFs).

Investors who want to invest in the Vanguard Total International Stock Market Index Fund often face a situation where they have to approximate it with the funds available in their employer sponsored plans, perhaps also using funds in other accounts, for example a Roth IRA. Thrift Savings Plan investors, for example, face this issue with the I Fund (based on MSCI EAFE index), which lacks emerging markets, small-cap stocks, and Canadian stocks.

Total international stock market index

This table shows Total International Stock Market Index Funds from two providers. The aim is to approximate these stock style boxes using available funds chosen from the two-fund and EAFE tables below.

Total International Stock Market index fund	Morningstar style box
100% Vanguard Total International Stock ETF (VXUS) (https://investor.vanguard.com/investment-products/etfs/profile/vxus) (0.05%) Updated 30 May 2021[1][note 1]	Value / Blend / Growth Large: 24 / 32 / 21 Mid: 6 / 8 / 5 Small: 1 / 2 / 1
100% Fidelity Total International Index Fund (FTIHX) (https://fundresearch.fidelity.com/mutual-funds/summary/31635V638) (0.06%) Updated 16 July 2021[1][note 1]	Value / Blend / Growth Large: 23 / 34 / 20 Mid: 6 / 8 / 5 Small: 1 / 2 / 1

Approximating total international stock with two funds

These examples match the composition of the total international stock market using two funds.

Match Total International Stock with two funds	Morningstar style box
90% Vanguard FTSE All-World ex-US ETF (VEU) (https://investor.vanguard.com/investment-products/etfs/profile/veu) (0.04%) 10% Vanguard FTSE All-World ex-US Small-Cap ETF (VSS) (https://investor.vanguard.com/investment-products/etfs/profile/vss) (0.08%) Updated 30 May 2021[1][note 1]	Value / Blend / Growth Large: 24 / 32 / 21 Mid: 6 / 8 / 5 Small: 1 / 2 / 1
90% Fidelity ZERO International Index Fund (FZILX) (https://fundresearch.fidelity.com/mutual-funds/summary/31635T609) (0.00%) 10% Vanguard FTSE All-World ex-US Small-Cap ETF (VSS) (https://investor.vanguard.com/investment-products/etfs/profile/vss) (0.08%) Updated 16 July 2021[1][note 1]	Value / Blend / Growth Large: 24 / 33 / 20 Mid: 5 / 9 / 5 Small: 1 / 2 / 1
75% Vanguard FTSE Developed Markets ETF (VEA) (https://investor.vanguard.com/investment-products/etfs/profile/vea) (0.03%) 25% Vanguard FTSE Emerging Markets ETF (VWO) (https://investor.vanguard.com/investment-products/etfs/profile/vwo) (0.07%) Updated 30 May 2021[1][note 1]	Value / Blend / Growth Large: 24 / 33 / 20 Mid: 6 / 8 / 5 Small: 1 / 2 / 1
75% SPDR Portfolio Developed World ex-US ETF (SPDW) (https://www.ssga.com/us/en/institutional/etfs/funds/spdr-portfolio-developed-world-ex-us-etf-spdw) (0.04%) 25% SPDR Portfolio Emerging Markets ETF (SPEM) (https://www.ssga.com/us/en/institutional/etfs/funds/spdr-portfolio-emerging-markets-etf-spem) (0.11%) Updated 14 July 2021[1][note 1]	Value / Blend / Growth Large: 24 / 34 / 21 Mid: 6 / 8 / 4 Small: 2 / 2 / 1

70% iShares Core MSCI International Developed Markets ETF (IDEV) (https://www.ishares.com/us/products/286762/ishares-core-msci-international-developed-markets-etf/) (0.05%)
30% iShares Core MSCI Emerging Markets ETF (IEMG) (https://www.ishares.com/us/products/244050/) (0.11%)

Updated 23 June 2021[1][note 1]

	Value	Blend	Growth
Large	23	34	20
Mid	6	9	5
Small	1	1	1

Developed and emerging market country classifications vary depending on the index provider.[2] For this reason, it is best if you use both funds from the same provider, as shown in the last three examples above.

Approximating total international stock with an EAFE fund

These examples were designed specifically if you need to use a Europe, Australasia and Far East (EAFE) index fund, such as the TSP I Fund or the Schwab International Index Fund (SWISX) (https://www.schwabassetmanagement.com/products/swisx). They show how you can combine an EAFE fund with other funds to match the composition of the total international stock market.

Match Total International Stock using an EAFE fund	Morningstar style box
58% MSCI EAFE Index Fund[3] 27% iShares Core MSCI Emerging Markets ETF (IEMG) (https://www.ishares.com/us/products/244050/) (0.11%) 10% Vanguard FTSE All-World ex-US Small-Cap ETF (VSS) (https://investor.vanguard.com/investment-products/etfs/profile/vss) (0.08%) 5% iShares MSCI Canada ETF (EWC) (https://www.ishares.com/us/products/239615/ishares-msci-canada-etf) (0.51%) Updated 30 May 2021[1][note 1]	Value / Blend / Growth Large: 24 / 34 / 21 Mid: 5 / 8 / 4 Small: 2 / 2 / 1
58% MSCI EAFE Index Fund[3] 23% Vanguard FTSE Emerging Markets ETF (VWO) (https://investor.vanguard.com/investment-products/etfs/profile/vwo) (0.07%) 10% Vanguard FTSE All-World ex-US Small-Cap ETF (VSS) (https://investor.vanguard.com/investment-products/etfs/profile/vss) (0.08%) 5% iShares MSCI Canada ETF (EWC) (https://www.ishares.com/us/products/239615/ishares-msci-canada-etf) (0.51%) 4% iShares MSCI South Korea ETF (EWY) (https://www.ishares.com/us/products/239681/ishares-msci-south-korea-capped-etf) (0.59%) Updated 30 May 2021[1][note 1]	Value / Blend / Growth Large: 24 / 34 / 21 Mid: 5 / 8 / 4 Small: 2 / 2 / 1

The MSCI EAFE index does not include:

- Emerging markets
- International small-caps
- Canadian stocks

You may want to use one of the two boxes depending on whether you have access to an MSCI or FTSE emerging markets index fund. MSCI classifies South Korea as an emerging market, while FTSE considers it a developed market;[4] this is why box 2 uses a separate South Korea index fund.

The blended expense ratios of the two boxes are 0.10% and 0.12% respectively, assuming that the I Fund (0.055%) (https://www.tsp.gov/funds-individual/i-fund/?tab=fees) takes the place of the MSCI EAFE index fund. You can reduce the expense ratios by using the Franklin FTSE Canada ETF (FLCA) (https://www.franklintempleton.com/investor/investments-and-solutions/investment-options/etfs/overview/26364/franklin-ftse-canada-etf/FLCA) (0.09%) and Franklin FTSE South Korea ETF (FLKR) (https://www.franklintempleton.com/investor/investments-and-solutions/investment-options/etfs/o

verview/26353/franklin-ftse-south-korea-etf-/FLKR) (0.09%). These are smaller and more lightly traded than their iShares equivalents, meaning a larger bid/ask spread, but they have much lower expense ratios.

To hold the portfolio in box 1 in the recommended percentage, using the Admiral Class share of the Vanguard FTSE All-World ex-US Small Cap Index Fund (VFSAX) (https://investor.vanguard.com/investment-products/mutual-funds/profile/vfsax), the minimum amount you would need for all four funds together is $30,000. This is because 10% of that amount equals the $3,000 fund minimum for VFSAX. Of that $30,000, at least $12,600 must be in the IRA to hold the three funds, with the remaining $17,400 in the I Fund.[note 2] You can avoid this by owning ETF share classes of the Vanguard FTSE All-World ex-US Small Cap Index.

Canadian investors

For Canadians, most Registered Retirement Savings Plans have a brokerage function that lets you hold securities based in the United States. Consider investing in Vanguard Total World Stock Index ETF (https://investor.vanguard.com/investment-products/etfs/profile/vt) which samples securities from throughout the entire investable world market. You could then use Vanguard MSCI Canada Index ETF (https://www.vanguardcanada.ca/individual/etfs/etfs-detail-overview.htm?portId=9554) and iShares S&P/TSX Completion Index Fund (https://ca.ishares.com/product_info/fund/overview/XMD.htm) to balance to your desired domestic exposure.

Methodology

All the above collections of funds were weighted in a manner to approximate the Morningstar Style Box, average market cap, and industry and regional weightings of the Vanguard Total International Stock ETF (VXUS) (0.08%) (https://investor.vanguard.com/investment-products/etfs/profile/vxus).[1][note 1]

The collection of funds making up the first box in the EAFE table is weighted in the following way:

- **Small-cap stocks** are weighted at 10% per Vanguard FTSE All-World ex-US Small-Cap Index.
- **Canadian stocks** are 6.6% of Vanguard's Total International Stock ETF[5] and 14.8% of Vanguard's FTSE All-World ex-US Small Cap ETF.[6] Given that Vanguard's international small cap fund overlaps the total international market fund, it was decided to weight the Canadian MSCI fund at 5% of the overall collection of funds (which is 6.6% - (14.8% × 10%)).
- **Emerging markets** (per MSCI) are 29.6% of Vanguard's Total International Stock ETF,[note 3] and are 28.3% of the Vanguard FTSE All-World ex-US Small-Cap ETF.[note 4] Given that these funds overlap, the MSCI Emerging Markets ETF was weighted at 27% (which is 29.6% - (28.3% × 10%)).
- **Europe, Australasia, and the Far East (EAFE)** is everything left over, which is 58%. The MSCI EAFE index fund is a proxy for the I Fund, Schwab International Index Fund (SWISX) or any MSCI EAFE fund.

Region and sector comparison

When Vanguard's International Total Stock ETF and the collection of funds were compared in Morningstar's X-Ray's different measurements, both were similar. For example, X-Ray compared the geographic regions as (rounded to the nearest 1%):

Region	Vanguard Total International Stock ETF	I Fund + MSCI Canada + FTSE Emerging Markets + MSCI South Korea + Small Cap	I Fund + MSCI Canada + MSCI Emerging Markets + Small Cap
North America	7	7	7
Latin America	2	2	2
United Kingdom	9	9	9
Europe Developed	29	31	31
Europe Emerging	1	1	1
Africa/Middle East	3	3	3
Japan	16	15	15
Australasia	5	5	5
Asia Developed	12	12	12
Asia Emerging	16	15	15
Not Classified	0	0	0

When X-Ray compares the stock sectors, the numbers (rounded to the nearest 1%) are:

Sector	Vanguard Total International Stock ETF	I Fund + MSCI Canada + FTSE Emerging Markets + MSCI South Korea + Small Cap	I Fund + MSCI Canada + MSCI Emerging Markets + Small Cap
Cyclical	**42**	**42**	**42**
Basic Materials	8	9	9
Consumer Cyclical	12	12	12
Financial Services	18	18	18
Real Estate	4	4	4
Sensitive	**38**	**37**	**37**
Communication Services	7	7	7
Energy	4	4	4
Industrials	13	13	13
Technology	13	13	13
Defensive	**20**	**21**	**21**
Consumer Defensive	8	8	8
Healthcare	9	9	9
Utilities	3	3	3

These percentages were last updated on May 30, 2021.

Notes

1. Morningstar's *Instant X-Ray* portfolio analysis tool has been discontinued effective April 17, 2025. Source: "Upcoming changes to our membership offerings, tools, and features" (https://global.morningstar.com/en-gb/updates/upcoming-changes-our-membership-offerings-tools-features). *Morningstar*. April 15, 2025. Retrieved March 10, 2026. Forum member LadyGeek has developed an alternative Equity style box calculator (https://docs.google.com/spreadsheets/d/113HpxXv5NrKWDXpv_xGNIVeOQk3giuRD/edit?pli=1&gid=63194195#gid=63194195) spreadsheet. Discussion and support is in Bogleheads forum topic: "Wiki - Morningstar's free Instant X-ray tool no longer available - Alternatives?" (https://www.bogleheads.org/forum/viewtopic.php?t=469099), LadyGeek. March 08, 2026.
2. For an older version of the weightings and price calculation, see the Vanguard Total International Stock Market replication with I Fund (https://docs.google.com/spreadsheet/ccc?key=0AueLhNpKZa4XdDdDVEFiZlJKWHlEUFJFVzJBaVlqakE) Google Spreadsheet.
3. 29.6% = 25.5% FTSE emerging markets + 4.1% South Korea.[5]

4. 28.3% = 23.4% FTSE emerging markets + 4.9% South Korea.[6]

See also

- Approximating total stock market

References

1. Morningstar X-Ray
2. "Is South Korea crowding your emerging markets allocation?" (https://www.spglobal.com/en/research-insights/articles/is-south-korea-crowding-your-emerging-markets-allocation). S&P Global. November 24, 2020.
3. Approximated with Schwab International Index Fund (SWISX) (0.06%) (https://www.schwabassetmanagement.com/products/swisx), May 30, 2021.
4. "South Korea: Developed or Emerging?" (https://www.ftserussell.com/blogs/south-korea-developed-or-emerging). FTSE Russell. February 19, 2014.
5. "Vanguard Total International Stock ETF (VXUS) Portfolio & Management" (https://investor.vanguard.com/investment-products/etfs/profile/vxus). Vanguard. April 30, 2021.
6. "Vanguard FTSE All-World ex-US Small-Cap ETF (VSS) Portfolio & Management" (https://investor.vanguard.com/investment-products/etfs/profile/vss). Vanguard. April 30, 2021.

External links

- Bogleheads forum topic: "Wiki - Approximating total international stock market - update for TSP" (https://www.bogleheads.org/forum/viewtopic.php?t=343859), LadyGeek. March 22, 2021

Retrieved from "https://www.bogleheads.org/w/index.php?title=Approximating_total_international_stock_market&oldid=90824"